

Khaled Ibrahim Ahmed

Software Developer

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📅 1996/11/23 🛡 Exemption 🔗 LinkedIn 🐙 Github

Summary

Experienced .NET Developer with strong skills in building scalable applications and implementing DevOps practices. Adept at developing backend services, migrating legacy systems, and optimizing system performance and security. Focused on delivering efficient and reliable solutions

Professional Experience

Backend Engineer @AlAhly Momkn For E-Payments

2024/03 – present

- **Role:** Develop backend services for financial applications, including APIs and transaction management. Ensure system performance and security.
- **Technologies:** C#, .NET Core, SQL Server, PostgreSQL, Kafka, Docker, Git.
- **Achievement:** Implemented a Distributed Logging System to improve system monitoring and debugging.

Full Stack Developer @ Alqemam qTech

2023/04 – 2024/03

- **Led** the migration of a legacy WebForms application to .NET Core, significantly enhancing scalability and performance.
- **Executed** code refactoring and restructuring, aligning with .NET Core best practices.

Financial Analysis @seliz

2020 – 2022

Financial Accountant @seliz

2019 – 2020

Private Tutor for Macroeconomics, Microeconomics and Monetary Policy

2018 – 2019

Projects

E-Commerce Microservices

Developed a comprehensive e-commerce platform leveraging microservices architecture, showcasing expertise in modern distributed systems design and implementation.

Technologies Used:

- **Backend:** .NET Core, gRPC, Serilog , MediatR, FluentValidation, AutoMapper, MassTransit, Polly, IdentityServer4
- **ORM:** Entity Framework Core, Dapper
- **Message Brokers:** RabbitMQ
- **Infrastructure Monitoring:** ELK Stack (Elasticsearch, Kibana), Portainer
- **Database:** SQL Server, PostgreSQL, Redis, MongoDB
- **API Gateway:** Ocelot
- **Tools:** Docker, Git, Visual Studio, Postman
- **Patterns and Practices:** Clean Architecture, Repository Pattern, CQRS, Mediator, Event Sourcing, Database-Per-Service, Dependency Injection, Domain-Driven Design (DDD)

Architecture Highlights:

- Designed and implemented a microservices-based system with dedicated services for Catalog, Basket, Discount, and Ordering.
- Employed API Gateways for efficient routing and load balancing.
- Integrated event-driven architecture using RabbitMQ for robust inter-service communication.

Key Features:

- Containerized microservices using Docker, ensuring consistency across environments.
- Orchestrated the system with Kubernetes, enhancing scalability and resilience.
- Implemented Pub/Sub patterns for seamless inter-service communication.
- Utilized Elasticsearch for efficient and scalable data querying.

DevOps & Monitoring:

- Deployed comprehensive logging and monitoring solutions using the ELK Stack, ensuring continuous system health and performance tracking.

StreamNet

CQRS & Event Sourcing and Transactional OutBox Pattern with .NET Microservices

Led the design and implementation of a robust .NET microservices architecture, demonstrating CQRS (Command Query Responsibility Segregation) and Event Sourcing patterns without external frameworks like MediatR. The project emphasizes Domain-Driven Design (DDD) and includes advanced features like the Transactional Outbox Pattern to ensure consistency in distributed systems.

Key Contributions:

- **CQRS & Event Sourcing:** Implemented command and event handling, aggregate state management, and event versioning.
- **Mediator Pattern:** Developed custom command and query dispatchers using the mediator pattern.
- **Event Store & Replay:** Utilized MongoDB for event storage and implemented event replay to rebuild aggregate states and read databases.
- **Message Bus Integration:** Integrated Apache Kafka as a message bus for reliable inter-service communication.
- **Transactional Outbox Pattern:** Ensured atomicity between database operations and Kafka message publishing, enhancing reliability.
- **Optimistic Concurrency Control:** Applied strategies to manage concurrent modifications in a distributed environment.
- **Database-Per-Service Pattern:** Designed independent database management for each microservice, supporting scalability and isolation.
- **Docker Containerization:** Containerized microservices for consistent development and deployment environments.

Technologies: .NET Core, C#, MongoDB, Microsoft SQL Server, Apache Kafka, Docker, Entity Framework Core, Domain-Driven Design (DDD)

Centralized Logging System

Developed a flexible Centralized Logging System for managing and analyzing logs across distributed services. The system handles log data efficiently through modular components, integrating Serilog, RabbitMQ, and Elasticsearch.

Key Features:

- **Dynamic Configuration:** Easily configurable components for scalability.
- **Serilog Integration:** Set up Serilog for robust logging.
- **RabbitMQ & Elasticsearch:** Managed logs through RabbitMQ and indexed them in Elasticsearch for effective querying.

Technologies Used:

- **Backend:** .NET Core, Serilog, RabbitMQ, Elasticsearch
- **Tools:** Docker, Git

Impact:

- Enhanced log management and analysis capabilities across services.

MimiSharp

MimiSharp - Custom Lightweight Web Server Framework

Developed MimiSharp, a minimal web server framework in C# for building web applications with routing, middleware, and serialization.

Key Features:

- **Async HTTP Server**
- **Flexible Routing**
- **Built-in Middleware:** Error handling, logging, rate limiting
- **Serialization:** JSON, XML, HTML
- **Extensible:** Custom middleware and route grouping

EziOrm

EziOrm is a lightweight, custom Object-Relational Mapping (ORM) framework for .NET applications. It provides an easy-to-use interface for database operations while offering flexibility and control over your data access layer.

Injectify

Injectify is a lightweight, flexible Dependency Injection (DI) container for .NET applications. It provides a powerful way to manage dependencies in your projects, supporting constructor injection, property injection, and method injection

Education

Fullstack developer using .net, Information Technology Institute (ITI) - Ministry of Communication and Information Technology (MCIT) , Egypt.

2022 – 2023

Bachelor of Commerce, Economic Department, Zagazig University, Egypt

2015 – 2019

GPA: 84/100 Rank : the third

Graduation Project : Excellent

Skills

Programming Languages : (C, C++, C# ,goLang)

Frameworks (NET Core, ASP.NET  Core MVC)

Object Relation Mapper: (Entity Framework Core, Dapper)

API Development: (ASP.NET  API, gRPC)

Database : (SQL Server, PostgreSQL, MongoDB, Redis)

Message Brokers: (RabbitMQ, Kafka)

Monitoring & Logging: (ELK Stack (Elasticsearch, Kibana), Serilog)

CI/CD: (Azure DevOps)

Containerization: (Docker)

Web Technologies: (HTML5, CSS3, JavaScript, TypeScript, Angular)

Organizations

Resala Charity Organization · Zagazig

2015 – 2020

Assistant

Fundraising , Orphan Sponsorship